

# NTIRE 2025 Challenge on Low Light Image Enhancement: Methods and Results

|                    |                   |                             |                         |                              |                     |                 |        |
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## Abstract

*This paper presents a comprehensive review of the NTIRE 2025 Low-Light Image Enhancement (LLIE) Challenge, highlighting the proposed solutions and final outcomes. The objective of the challenge is to identify effective networks capable of producing brighter, clearer, and visually compelling images under diverse and challenging conditions. A remarkable total of 762 participants registered for the competition, with 28 teams ultimately submitting valid entries. This paper thoroughly evaluates the state-of-the-art advancements in LLIE, showcasing the significant progress.*

## 1. Introduction

Low-Light Image Enhancement (LLIE) aims to improve visibility and contrast across a wide range of low-light conditions. In addition to enhancing brightness, LLIE seeks to address issues such as noise, artifacts, and color distortion,

which are prevalent in dark scenes or arise during the illumination correction process.

Building upon the success of the NTIRE 2024 LLIE Challenge [35], we launched a new iteration at the NTIRE 2025 workshop. The 2025 challenge continues to encourage innovation by proposing solutions that significantly enhance image quality under complex low-light scenarios.

The goals of the challenge are threefold: (1) to drive research progress in the field of LLIE, (2) to enable systematic comparison of emerging methodologies, and (3) to provide a platform for academic and industrial participants to exchange ideas and explore potential collaborations.

Following a similar setup to the NTIRE 2024 edition [35], the dataset comprises a diverse set of scenarios under varying lighting conditions, including dim scenes, severe darkness, backlighting, non-uniform illumination, and both indoor and outdoor night scenes, with image resolutions reaching 4K and beyond. The dataset includes 219 training scenes, along with 46 for validation and 30 for testing. Ground-truth (GT) images for both the validation and testing sets were kept hidden from participants. Detailed dataset specifications will be published in future work.

This challenge is one of the NTIRE 2025 Workshop associated challenges on: ambient lighting normalization [46], reflection removal in the wild [51], shadow removal [45], event-based image deblurring [43], image de-

\* X. Liu, Z. Wu, H. Yan, F. Vasluianu, B. Ren, Y. Zhang, S. Gu, L. Zhang, C. Zhu and R. Timofte were the challenge organizers, while the other authors participated in the challenge. Each team described its own method in the report, shortened by the organizers to meet 8 page criteria. Appendix A contains the teams, affiliations and architectures if available. NTIRE 2025 webpage: <https://cvlai.net/ntire/2025>. Code: <https://github.com/AVC2-UESTC/NTIRE2025-LLIE>.

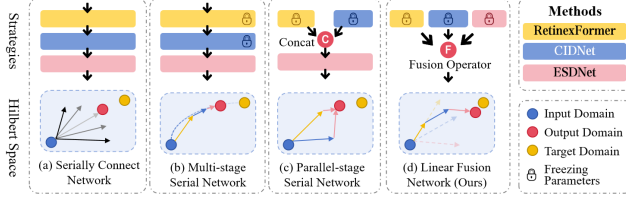


Figure 1. Architecture of winner solution

noising [44], XGC quality assessment [34], UGC video enhancement [41], night photography rendering [14], image super-resolution (x4) [9], real-world face restoration [10], efficient super-resolution [40], HR depth estimation [55], efficient burst HDR and restoration [28], cross-domain few-shot object detection [17], short-form UGC video quality assessment and enhancement [31, 32], text to image generation model quality assessment [19], day and night rain-drop removal for dual-focused images [30], video quality enhancement for video conferencing [22], low light image enhancement, light field super-resolution [48], restore any image model (RAIM) in the wild [33], raw restoration and super-resolution [11] and raw reconstruction from RGB on smartphones [12].

## 2. Tracks and Competition

**Ranking criteria.** To evaluate the submissions, we use conventional metrics such as PSNR, SSIM, LPIPS, and NIQE. As shown in Tab. 1, the “Final Rank” is a composite metric derived from a weighted combination of PSNR (50%), SSIM (50%), LPIPS (40%), and NIQE (20%).

**Challenge phases.** (1) *Development and validation phase:* Participants were provided with 219 training image pairs and 46 validation inputs from our custom dataset. The ground-truth images for the validation set were not shared. Participants could submit their enhanced results to an evaluation server, which computed PSNR and SSIM scores and provided real-time feedback. (2) *Testing phase:* Participants received 30 low-light test images, again without access to the corresponding ground-truth images. Submissions, including enhanced results, accompanying code, and a fact-sheet, were uploaded to the Codalab evaluation server and shared with the organizers. The organizers verified and the final results. Top-performing teams were required to submit training scripts to ensure reproducibility.

## 3. Challenge Methods and Teams

The results of the low light enhancement challenge are detailed in Tab. 1, which evaluates and ranks the performances of 28 teams. One team (JHC-Info) fails to provide the checkpoint for reproducibility within the competition period, hence excluded ranking. Some others team provide all the fact sheet but didnt participate in the challenge report. These works are highlight by Gray

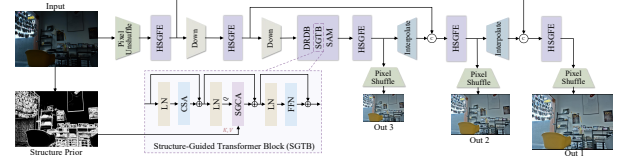


Figure 2. Architecture of 2nd place solution

### 3.1. NWPU-HVI

**Description:** We propose FusionNet, as shown in Fig. 1, a hybrid framework combining three complementary methods: ESDNet [54] for local feature processing, Retinexformer [6] for long-range dependencies, and CIDNet [50] utilizing the HVI color space. FusionNet introduces four fusion strategies: 1) Serial network (single-stage, risks instability); 2) Multi-stage serial training (frozen parameters, slower convergence); 3) Parallel training followed by serial enhancement (inefficient); 4) Linear fusion with fully parallel execution with weights:  $\mathbf{I}_{HQ} = \sum_{i=1}^n k_i \mathbf{F}_i(\mathbf{I}_{LQ})$

Each method defines a unique mapping in Hilbert space, and optimal fusion maximizes projection in the target subspace, enhancing domain generalization.

**Implementation:** The model is implemented in PyTorch with the Adam optimizer and cosine annealing schedule. CIDNet/ESDNet/Retinexformer is trained on  $1024^2/1600^2/2000^2$  patches for 90k/100k/180k iterations, respectively, all with a batch size 1.

### 3.2. Imagine

**Description:** We propose SG-LLIE, as shown in Fig. 2, a multi-scale CNN-Transformer hybrid network based on UNet architecture. The Hybrid Structure-Guided Feature Extractor (HSGFE) employs structural cues to preserve fine details. Down-sampling is achieved with “PixelUnshuffle” and convolutional layers, while up-sampling uses “PixelShuffle” or “Interpolate”. Skip connections maintain spatial coherence. The Color Invariant Convolution (CICov) extracts illumination-invariant priors, and the Structure-Guided Transformer Block (SGTB) modulates learning with Channel-wise Self-Attention (CSA), Structure-Guided Cross Attention (SGCA), and Feed-Forward Networks (FFN). The model employs Dilated Residual Dense Blocks (DRDB) and Semantic-Aligned Scale-Aware Modules (SAM) for hybrid local and long-range feature learning. Training samples are classified by illumination levels, with adjustment factors applied. A self-ensemble strategy further enhances performance. The total loss is a combination of combines Charbonnier loss [26], perceptual loss [27], and Multi-Scale SSIM loss [60] with a weighting of 1, 0.01 and 0.4, respectively.

**Implementation:** The model is trained on the NTIRE 2025 dataset with a patch size of  $1600^2$ , batch size of 1, initial learning rate of  $2 \times 10^{-5}$ , and 60,000 iterations with a cyclic cosine annealing schedule. The adjustment layer is trained with cross-entropy loss and softmax activation. The net-

Table 1. Evaluation and Rankings in the NTIRE 2025 Low Light Image Enhancement Challenge. “Rank”s indicate the respective standings of participants based on their performance in different metrics on the challenge’s test dataset. “Final Rank” represents a composite metric, derived from a weighted sum of 0.5, 0.5, 0.4, 0.2, respectively.

| Team             | PSNR  | SSIM  | LPIPS | NIQE  | Rank PSNR | Rank SSIM | Rank LPIPS | Rank NIQE | Final Rank |
|------------------|-------|-------|-------|-------|-----------|-----------|------------|-----------|------------|
| NWPU-HV1         | 26.24 | 0.861 | 0.128 | 10.95 | 2         | 2         | 7          | 11        | 1          |
| Imagine          | 26.35 | 0.858 | 0.133 | 11.81 | 1         | 3         | 9          | 23        | 2          |
| pengpeng-yu      | 25.85 | 0.858 | 0.134 | 11.29 | 4         | 3         | 11         | 16        | 3          |
| DAVIS-K          | 25.14 | 0.863 | 0.127 | 10.58 | 14        | 1         | 6          | 9         | 4          |
| SoloMan          | 25.80 | 0.856 | 0.13  | 11.49 | 5         | 6         | 8          | 19        | 5          |
| Smartdsp         | 25.47 | 0.848 | 0.12  | 10.53 | 11        | 12        | 3          | 8         | 6          |
| Smart210         | 26.15 | 0.855 | 0.137 | 11.52 | 3         | 7         | 14         | 20        | 7          |
| WHU-MVP          | 25.76 | 0.855 | 0.138 | 11.21 | 7         | 7         | 15         | 13        | 8          |
| BUPTMM           | 25.67 | 0.855 | 0.137 | 11.28 | 8         | 7         | 14         | 14        | 9          |
| NJUPT-IPR        | 25.01 | 0.848 | 0.122 | 10.15 | 15        | 12        | 5          | 3         | 10         |
| SYSU-FVL-T2      | 25.65 | 0.857 | 0.135 | 11.59 | 10        | 5         | 12         | 22        | 11         |
| KLETech-CEVI     | 25.66 | 0.854 | 0.134 | 11.55 | 9         | 10        | 11         | 21        | 12         |
| Ensemble-KNights | 25.77 | 0.849 | 0.139 | 11.47 | 6         | 11        | 16         | 18        | 13         |
| MRT-LLIE         | 24.52 | 0.833 | 0.117 | 10.23 | 19        | 18        | 2          | 4         | 14         |
| SynLLIE          | 24.01 | 0.84  | 0.117 | 10.37 | 22        | 15        | 2          | 5         | 15         |
| Cidaut AI        | 25.45 | 0.839 | 0.144 | 10.45 | 12        | 16        | 17         | 7         | 16         |
| Huabujianye      | 25.15 | 0.845 | 0.157 | 11.17 | 13        | 14        | 20         | 12        | 17         |
| no way no lay    | 24.64 | 0.839 | 0.154 | 11.32 | 17        | 16        | 18         | 17        | 18         |
| Lux Themps       | 22.27 | 0.822 | 0.122 | 10.39 | 27        | 21        | 5          | 6         | 19         |
| PSU_team         | 24.86 | 0.824 | 0.176 | 10.95 | 16        | 20        | 21         | 10        | 20         |
| hfut-lvgroup     | 24.54 | 0.832 | 0.157 | 11.29 | 18        | 19        | 20         | 15        | 21         |
| ImageLab         | 23.87 | 0.816 | 0.191 | 9.68  | 23        | 22        | 22         | 1         | 22         |
| AVC2             | 24.02 | 0.816 | 0.196 | 12.88 | 21        | 22        | 23         | 28        | 23         |
| LR-LL            | 24.22 | 0.816 | 0.236 | 12.10 | 20        | 22        | 27         | 25        | 24         |
| X-L              | 23.49 | 0.803 | 0.212 | 12.86 | 25        | 26        | 25         | 27        | 25         |
| Team_IITRPR      | 23.50 | 0.803 | 0.212 | 12.86 | 25        | 26        | 25         | 27        | 26         |
| CV-SVNIT         | 16.85 | 0.565 | 0.427 | 12.29 | 28        | 28        | 28         | 26        | 27         |
| JHC-Info         | 23.38 | 0.803 | 0.203 | 9.85  | 26        | 26        | 24         | 2         | -          |

work is fine-tuned on the NTIRE 2024 and 2025 datasets, with a learning rate of  $2 \times 10^{-7}$ .

### 3.3. pengpeng-yu

**Description:** We adopted and extended the EDSNet [54] implementation developed by the SYSU-FVL-T2 team in the NTIRE 2024 edition [35]. Our method builds upon an encoder-decoder architecture with three feature scales and skip connections. At each scale, Semantic-Aligned Scale-Aware Modules are stacked to enhance the model’s ability to process multi-scale features. The loss function integrates the Charbonnier loss [26] and the perceptual VGG19 loss [27]. To mitigate semantic degradation during downsampling, we enable anti-aliasing in the downsampling operations within the SAM modules and the perceptual loss computation. For inference, we adopt a self-ensembling strategy, applying geometric transformations to the input and averaging the corresponding predictions.

**Implementation:** We train the model for 150,000 iterations using the Adam optimizer on an NVIDIA RTX 4090 GPU, starting from the pretrained SYSU-FVL-T2 model weights with a resolution of 1600 and an initial learning rate of 0.0002, which is scheduled using cyclic cosine annealing. Inference is conducted at the original resolution.

### 3.4. DAVIS-K

**Description:** We propose a two-stage U-shaped Transformer, as illustrated in Fig. 3. The model comprises an *Enhancement Module* and a *Refinement Module*. The Enhancement Module contains two branches. The first branch adopts Restormer [56], a hierarchical U-Net with skip connections and Transformer blocks that increase in depth from top to bottom. It employs Multi-Dconv Head Transposed Attention for efficient channel-wise feature interaction. Input images are downsampled before processing and upsampled via PixelShuffle.

The second branch is designed for transfer learning. It uses the first three layers of a pre-trained ConvNeXt [36] as the encoder to leverage prior knowledge, followed by a lightweight decoder [61] for gradual upsampling. The Refinement Module, sharing the same architecture as the first branch, further enhances color and texture details.

**Implementation:** The model is trained using the Adam optimizer ( $\beta_1 = 0.9$ ,  $\beta_2 = 0.999$ ) for 600K iterations with a cosine-annealed learning rate from  $2 \times 10^{-4}$  to  $1 \times 10^{-6}$ . We use a batch size of 8 and randomly crop  $384 \times 384$  patches with rotation and flip augmentations. Training is conducted in PyTorch on an RTX 4090 GPU. This work is supported by the Technology Development Program (RS-2024-00469833), funded by the Ministry of SMEs and Star-

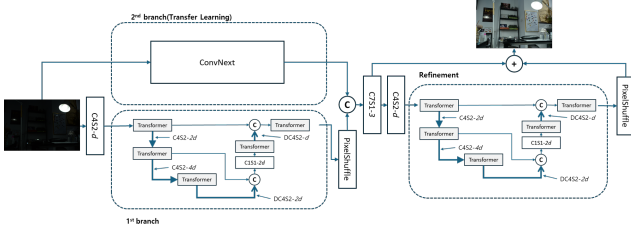


Figure 3. Architecture of Team DAVIS-K

tups (MSS, Korea).

### 3.5. SoloMan

**Description:** We propose **ESDNet-Twins**, a twin-network architecture employing a complementary ensemble strategy for low-light image enhancement. The first network, *Twin1*, follows the multi-scale design of ESDNet [54], which already delivers strong performance. However, it struggles in extremely over-/under-exposed regions and lacks in detail preservation and edge sharpness.

To address these limitations while retaining its strengths, we introduce *Twin2*—a lightweight counterpart with improved DB, RDB, and SAM blocks, and a novel multi-scale **Feature Full Connect Block** between the encoder and decoder. This design enables *Twin2* to handle higher-resolution images more efficiently. Instead of merging both networks into a single model—which could lead to overfitting due to the small dataset—we train them independently and apply an ensemble strategy, as shown in Fig. 4. We optimize the model using a multi-VGG perceptual loss:

$$\mathcal{L} = \mathcal{L}_{\text{VGG}} + \mathcal{L}_{\text{CB}}, \quad (1)$$

where  $\mathcal{L}_{\text{VGG}}$  denotes perceptual loss using a pretrained VGG16, and  $\mathcal{L}_{\text{CB}}$  is the Charbonnier loss.

**Implementation:** ESDNet-Twins is implemented in PyTorch and trained on an NVIDIA A6000 (48GB). *Twin1* is trained from scratch for 150K iterations using Adam. Training is staged with decreasing batch sizes  $\{8, 4, 4, 2, 2, 1\}$  and corresponding patch sizes  $\{720, 1024, 1024, 1280, 1600\}$ . *Twin2*, being more efficient, follows the same setup but with larger patches  $\{720, 1024, 1024, 1600, 1800\}$ . Both models use an initial learning rate of  $2 \times 10^{-4}$ , scheduled with cyclic cosine annealing.

### 3.6. Smartdsp (Excluded from Report)

### 3.7. Smart210

**Description:** Since PSNR and SSIM are the primary evaluation metrics, we adopt ESDNet [54]—a strong SOTA baseline in 2024—as our foundation. ESDNet employs an encoder-decoder structure with Dilated Residual Dense Blocks for feature extraction and Semantic-Aligned Scale-Aware Modules (SAM) for multi-scale fusion. As shown in Fig. 5, we replace the Dilated Residual Dense Block

with the SimPFblock [47], which is based on the NAF-Block [7] and incorporates the parameter-free attention mechanism SimAM [52], illustrated in Fig. 6. SimPFblock reduces multiplication operations—beneficial for low-light inputs where low pixel values may cause vanishing gradients—making it well-suited for this task. Experiments confirm that SimPFblock improves performance in PSNR and SSIM.

**Implementation:** Our method is implemented in Python 3.8 using PyTorch. Following the progressive training strategy in SYSU-FVL-T2 [35], we train for 250K iterations using the Adam optimizer ( $\beta_1 = 0.99$ ,  $\beta_2 = 0.999$ ) with an initial learning rate of 0.0002, decayed via cyclic cosine annealing. Final predictions are generated by linearly combining outputs from checkpoints at 150K, 200K, and 205K iterations. We also adopt the self-ensemble strategy from Retinexformer [6], which consistently boosts PSNR.

### 3.8. WHU-MVP

**Description:** We adopt ESDNet [54] as the backbone for LLIE. Given the high-resolution nature of low-light images, computing global attention directly can be computationally expensive. To address this, inspired by Transformer-based approaches [56], we introduce a parallel restoration branch operating on  $4 \times$  downsampled inputs to perform coarse enhancement. The output is then refined by ESDNet at the original resolution. This coarse-to-fine strategy enhances adaptability to high-resolution inputs and improves overall restoration quality. Extensive experiments validate the effectiveness of this design.

**Implementation:** Training is performed exclusively on the official competition dataset using four NVIDIA RTX 4090 GPUs. A progressive training scheme is employed with increasing patch sizes  $\{1024, 1280, 1280, 1600, 1920\}$  and corresponding batch sizes  $\{4, 4, 2, 2, 1\}$ , over  $\{46\text{K}, 32\text{K}, 24\text{K}, 18\text{K}, 30\text{K}\}$  iterations, respectively. The initial learning rate is  $2 \times 10^{-4}$ , scheduled via cyclic cosine annealing. For inference, images under  $3000 \times 3000$  are processed directly, while larger images are split, processed in segments, and reassembled to reduce GPU memory usage.

### 3.9. BUPTMM (Excluded from Report)

### 3.10. NJUPT-IPR

**Description:** We adopt an enhanced version of our previous work IAG-CoFlow [20] for this competition. IAG-CoFlow is a normalizing flow-based method for low-light image enhancement, comprising a conditional generator and a complete flow module. IAG is a U-shaped Transformer network built with IIZAT (inter-/intra-channel and zeromap attention Transformer) for downsampling and IIAT (inter-/intra-channel attention Transformer) for up-sampling. IIAT models inter- and intra-channel attention independently, while IIZAT further integrates zeromap at-



tention in parallel. CoFlow introduces three novel invertible transformations—linear injector, conditional linear coupling, and unconditional linear coupling—guided by a cross-attention network to model affine transformations conditioned on features. During training, CoFlow takes  $I_H$ ,  $Ft1$ ,  $Ft2$ , and  $Ft3$  as inputs to produce latent variables  $z = \Phi(I_H; I_L)$ , which are mapped to  $\mathcal{N}(0, 1)$ . In inference,  $z$  and feature maps generate the enhanced image  $I_E$  from the low-light input  $I_L$ .

**Implementation:** The method is implemented in PyTorch and trained with the Adam optimizer for 150,000 iterations. The learning rate starts at  $2 \times 10^{-4}$  and is halved at 25%, 50%, 70%, and 80% of training. Training uses  $896 \times 896$  random crops with a batch size of 1. At test time,  $2000 \times 3000$  images are processed directly, while  $4000 \times 6000$  images are split into four tiles and merged post-enhancement. This work is supported by the National Natural Science Foundation of China (Grant 62272240).

### 3.11. SYSU-FVL-T2

**Description:** We propose a low-light image enhancement method based on ESDNet-L [54], as shown in Fig. 7. The model adopts an encoder-decoder architecture with three feature scales, connected via skip connections. Multi-scale features are generated using Lanczos3 interpolation. At each scale, two stacked Semantic-Aligned Scale-Aware Modules (SAM) are used to enhance the model’s ability to handle scale variations. Each SAM integrates a pyramid context extraction module and a cross-scale dynamic fusion module for selective multi-scale fusion. The total loss  $L_{total}$  is defined for the outputs at three scales with a combination of Charbonnier loss [26], perceptual loss [27], and Multi-Scale SSIM loss [60], and the color loss [58], with a weighting of 1, 0.04, 1 and 1, respectively.

**Implementation:** The method is implemented in Python 3.8 and trained on an NVIDIA RTX A6000 (49GB). Following the progressive training strategy of MIRNet-v2 [57], we train the model from scratch for 156,000 iterations using the Adam optimizer [24]. Initially, we use a batch size of 8 with  $720 \times 720$  patches, gradually adjusting to batch sizes of 4, 4, 2, 2, 1, and 2, and patch sizes of 1024, 1024, 1280, 1280, 1600, and 1440 at respective iteration stages (46k, 32k, 24k, 18k, 18k, and 6k). The learning rate starts at  $2 \times 10^{-4}$  and follows a cyclic cosine annealing schedule [37]. During inference, the full-resolution image is processed directly with a batch size of 1.

### 3.12. KLETech-CEVI

**Description:** We propose ESDNet+, as shown in Fig. 8, an enhanced version of ESDNet [54], for efficient low-light enhancement of ultra-high-definition (4K) images. The method utilizes a single-stage pipeline with a pre-processing head, encoder-decoder architecture, and inter-

mediate supervision. Key components like Dilated Residual Dense Blocks (DRDB) and Semantic-Aligned Scale-Aware Modules (SAM) are retained, with the introduction of a novel Low-Light Enhanced Perceptual Loss. The pipeline begins by downsampling the input image and extracting features with a  $5 \times 5$  depth-wise convolution. These features are processed through a three-level encoder with DRDB and SAM for multi-scale feature fusion. The encoder’s output is upsampled in the decoder, with skip connections to preserve high-resolution details. The final output is a fully enhanced 4K image, with intermediate outputs supervised during training. The loss function combines perceptual, luminance, and edge-preserving losses to optimize performance in low-light conditions:

$$L_{ESDNet+} = \alpha \cdot L_{VGG} + \beta \cdot L_{Luminance} + \gamma \cdot L_{Edge}.$$

**Implementation:** Participants did not provide details.

### 3.13. Ensemble-KNights

**Description:** We propose the ENsemble Bayesian Enhancement Model (EN-BEM), an enhancement of BEM [21] that integrates Transformer and Mamba architectures. EN-BEM leverages BEM’s two-stage approach, where a Bayesian Neural Network (BNN) models one-to-many mappings in the first stage, and a Deterministic Neural Network (DNN) refines image details in the second stage. In EN-BEM, the backbone is replaced with either a Transformer or Mamba architecture, with outputs combined through internal ensembling. This ensemble method balances computational efficiency, noise suppression, and detail restoration, while addressing the one-to-many mapping issue in low-light enhancement. The probabilistic nature of the BNN enables EN-BEM to capture data uncertainty, making it robust in dynamic low-light conditions. The overall framework is shown in Fig. 9.

**Implementation:** The models are implemented in PyTorch and trained on the provided dataset without external data. Training and testing are performed on a single RTX 4090 GPU. The Adam optimizer is used with an initial learning rate of  $2 \times 10^{-4}$ , decaying to  $10^{-6}$  using a cosine annealing schedule. The first-stage model is trained for 300K iterations on  $1792 \times 1792$  inputs, and the second-stage model for 150K iterations on  $496 \times 496$  inputs, with a batch size of 8. During inference, images are processed at full resolution, with a batch size of 1. A self-ensemble technique is applied during testing.

### 3.14. MRT-LLIE

**Description:** We propose MRT, a novel Transformer network leveraging a new encoder-decoder scheme called the **Multi-scale Entanglement Scheme**. Inspired by [35] (Sec. 4.16), this scheme is tailored for Transformers to learn enhanced multiscale feature representations. Additionally, we

introduce a **Residual Multi-headed Self-Attention** mechanism to preserve details across network stages. The **Multi-stage Squeeze & Excite Fusion Block** [5] is incorporated in the post-attention step for improved feature extraction. The design of **MRT** is shown in Fig. 10.

**Implementation:** MRT is implemented in PyTorch, trained on the NTIRE25 dataset. The model is optimized using the Adam optimizer for 150k iterations, with an initial learning rate of  $2e-4$ , decaying via Cosine Annealing. Each iteration uses a batch of two  $704 \times 704$  randomly-cropped image patches with data augmentation (random flipping/rotation). We employ a hybrid loss function that captures pixel-level, multi-scale, and perceptual differences. Testing is conducted via standard inference, except for  $4000 \times 6000$  images, which are split into four  $4000 \times 1500$  images using pixel interleaving to manage resource constraints.

### 3.15. SynLLIE (Excluded from Report)

### 3.16. Cidaut AI

**Description:** We propose two original models: **FLOL** [4] and **DarkIR** [15], both utilizing Fourier frequency information and the NAFBlock [7] architecture. The architecture, as shown in Fig. 11, consists of two stages in each network: (1) an illumination stage that enhances the image to the optimal lightness, and (2) a Denoiser stage (FLOL) or Deblur stage (DarkIR) that refines the enhanced image by removing noise, blur, and imperfections from the first stage. For FLOL, we use the Semantic-Aligned Scale-Aware Modules (SAM) [54] loss, combining perceptual and distortion terms across multiple crop sizes to improve performance. The loss is defined as:

$$\mathcal{L} = \sum_{i=1}^3 (\mathcal{L}_1 + \mathcal{L}_{inter} + \lambda \mathcal{L}_{LPIPS}), \quad (2)$$

where  $\lambda = 0.1$ ,  $\mathcal{L}_1$  is the  $L_1$  loss (MAE),  $\mathcal{L}_{inter}$  is the  $L_1$  loss for the intermediate image,  $i$  is the hierarchical scale, and  $\mathcal{L}_{LPIPS}$  is the perceptual loss from the VGG19 model [42]. For DarkIR, the multisize sum loss is not implemented. Additional optimization details are provided in the respective papers [4, 15].

**Implementation:** Both models are implemented in PyTorch. For DarkIR, we use the Adam optimizer with weight decay  $1 \times 10^{-3}$  and a learning rate of  $1 \times 10^{-3}$ , following a Cosine Annealing schedule down to  $1 \times 10^{-7}$ . Training consists of 5 stages with epoch lengths [250, 150, 100, 50, 50], varying crop sizes [384, 720, 1024, 1280, 1280] and batch sizes [24, 8, 4, 4, 4], using 4 H100 GPUs for approximately 6 hours. For FLOL, the Adam optimizer is used with a learning rate of  $2 \times 10^{-4}$ , also following a Cosine Annealing schedule. The training comprises 9 stages with crop sizes [720, 1024, 1024, 1280, 1280, 1600, 2000, 2200, 2400] and batch sizes [8, 4, 4, 2, 2, 1, 1, 1, 1], trained on a

single NVIDIA GeForce RTX 4090 GPU for approximately 72 hours across 2000 epochs. Both models utilize random square crops ( $H = W$ ) and random vertical and horizontal flips for data augmentation.

### 3.17. D-RetinexMix

**Description:** We propose an efficient multiscale network architecture that adapts DiffLL-based [23] generated results to match the illumination conditions of the dataset. The enhanced outputs from the RetinexFormer [6] pre-trained model are fused using an objective evaluation metric to produce high-quality results. The entire process is implemented as an end-to-end training framework. Specifically, during training, DiffLL generates pre-enhanced results from low-light images in the training set. We then construct a U-shaped network with dual branches: Vmamba and convolution blocks. The pre-enhanced results and the original low-light images are concatenated along the channel dimension and fed into this network to generate the second enhanced output. Finally, the second enhanced results are evaluated against the outputs from the RetinexFormer pre-trained model using an objective metric to select the highest-quality images as the final output.

**Implementation:** Experiments are conducted on a single NVIDIA GeForce RTX 3090 GPU with 24GB of memory, training for 50k iterations with random horizontal and vertical flipping. The Adam optimizer is used with a learning rate of  $4 \times 10^{-4}$ , a patch size of  $512 \times 512$ , and a batch size of 8. The results from DiffLL and RetinexFormer are obtained using the pre-trained weights.

### 3.18. No Way No Lay (Excluded from Report)

### 3.19. Lux Thempis

**Description:** Incorporating semantic information from MobileNetV3 [8], our method, **SADe-ViT**, adapts illumination by distinguishing regions and objects. The architecture follows a U-shaped encoder-decoder ViT design (see Fig. 12), where Transformer blocks integrate segmentation maps into attention mechanisms, enhancing feature representation and illumination adjustment. A multi-headed self-attention mechanism highlights key regions, followed by element-wise multiplication with spatially adapted semantic features to ensure dimensional alignment. To improve computational efficiency, CNNs and fully connected layers in the FFN are replaced by Depthwise Separable Convolutions, as in [2, 3], resulting in a model with only 0.57M parameters. The FFN output (Eq. (3)) is normalized using Layer Normalization (LN).

$$\mathbf{F}'_{in} = \text{DSC}^2 \sum_{i=1}^2 \text{DSC}^2(\mathbf{F}_{in}), \quad \mathbf{F}'_{in} \in \mathbb{R}^{H \times W \times C}. \quad (3)$$

A hybrid loss function evaluates multiple aspects of the

generated images:

$$\mathcal{L} = \alpha\mathcal{L}_2 + \beta\mathcal{L}_{perc} + \gamma\mathcal{L}_{SSIM} \quad (4)$$

where  $\mathcal{L}_2$  is the mean squared error,  $\mathcal{L}_{perc}$  is the perceptual loss using VGG-19 [27], and  $\mathcal{L}_{SSIM}$  [49] is based on the Multi-Scale Structural Similarity Index.

**Implementation:** Implemented in PyTorch, training is optimized using the Adam optimizer with a cosine annealing learning rate schedule, starting at  $2 \times 10^{-4}$ , increasing to  $3 \times 10^{-4}$ , and gradually decaying to  $1 \times 10^{-6}$ . We train for 300k iterations on  $256 \times 256$  augmented patches from the LOL-v2 real [53] dataset, focusing on severe light-deficient images. For testing phase, Evaluation is performed on the competition’s test dataset, including images of sizes  $2000 \times 2992$  and  $4000 \times 6000$ . For the larger images, we apply a tile-based enhancement, splitting them into four tiles, enhancing each, and then reconstructing the full image.

### 3.20. PSU\_team

**Description:** We introduce **OptiMalDiff** that reformulates image denoising as an optimal transport problem. The approach models the transition from noisy to clean images using a Schrödinger Bridge-based diffusion process. The architecture, as shown in Fig. 13, comprises: 1) a hierarchical Swin Transformer backbone for efficient extraction of local and global features; 2) a Schrödinger Bridge Diffusion Module for learning forward and reverse stochastic mappings, and (3) a Multi-Scale Refinement Network (MRefNet) for progressively refining image details. Additionally, a PatchGAN discriminator is integrated for adversarial training to enhance realism.

**Implementation:** The model is trained from scratch using the Low Light Image Enhancement dataset, without pre-trained weights. We jointly optimize all modules with a composite loss function, combining diffusion loss, Sinkhorn-based optimal transport loss, multi-scale SSIM and  $L_1$  losses, and an adversarial loss. Training is conducted over 300 epochs with a batch size of 8, totaling 35,500 iterations per epoch.

### 3.21. hfut-lvgroup

**Description:** Our approach integrates Retinexformer [6] with a U-Net variant [16] using a simple yet effective model fusion strategy. Both models independently process the low-light image, with the final output obtained by weighted averaging. Retinexformer, grounded in Retinex theory and enhanced by the Transformer architecture, excels at global modeling, making it effective for complex lighting conditions. The U-Net variant combines the full-resolution features of FRC-Net [59] with classic U-Net layers, learning both spatial structure and semantic information through stacked residual blocks at various scales.

**Implementation:** The models were trained separately using the NTIRE 2025 challenge dataset:

- Retinexformer: Trained on two NVIDIA RTX 4090 GPUs with a batch size of 1 for 150,000 iterations. The learning rate started at  $1e-4$  for the first 80,000 iterations and decreased to  $1e-5$  for the remaining iterations.
- U-Net Variant: Trained on a single NVIDIA RTX 4090 GPU for 1,000 epochs, with a learning rate starting at  $1e-4$  and gradually reduced to  $1e-5$  for the first 500 epochs, then maintained at  $2e-5$  for the next 500 epochs.

Both models used the Adam optimizer and L1 loss, with random cropping of images into  $1400 \times 1400$  patches to improve learning efficiency.

### 3.22. ImageLab

**Description:** The Lightweight Self-Calibrated Pixel-Attentive Network for Low-Light Image Enhancement (LLIE-Net), as shown in Fig. 15, enhances low-light images using two inputs: an RGB image and its HSV-derived pixel-level features. The RGB input is downsampled and processed through five Self-Calibrated Pixel Attention (SCPA) blocks [62] for noise suppression and feature recalibration. These features are upsampled and fused with HSV features for color-aware enhancement [39]. A multi-stage encoder incorporating Residual Dense Attention (RDA) [38] and NAFBlockSR [13] modules aggregates features and captures context. The decoder mirrors the encoder with up-sampling and skip connections, using RDA blocks to refine textures. Auxiliary branches process the original input with attention mechanisms to preserve details and avoid over-smoothing. Outputs from the decoder, auxiliary branch, and a shallow pathway are fused with the original input to produce the final enhanced image.

**Implementation:** LLIE-Net was trained on an NVIDIA Tesla P100 (16GB RAM) using TensorFlow Keras. It was trained on 4,281 patches ( $400 \times 400 \times 3$ ) with random augmentations and validated on 755 patches. The Adam optimizer was used with a learning rate decaying from 0.001 to 0.00001 over 250 epochs.

### 3.23. AVC2

**Description:** We propose MobileIE, an efficient model for low-light image enhancement that balances parameters, speed, and performance. The model follows a simple design, utilizing basic operations in a streamlined topology (see Fig. 16). During training, low-light images are processed through MBRConv  $5 \times 5$  and PReLU to extract shallow features. These features are then passed through two modules combining MBRConv  $3 \times 3$  and FST for further deep feature learning. An Attention module focuses on important regions, followed by MBRConv  $3 \times 3$  for fine processing to produce the final result. Inference uses reparameterized MBRConv layers.

**Implementation:** The model is implemented in PyTorch and tested on an RTX 3090 GPU. Optimization is done with the Adam optimizer and a cosine annealing learning rate schedule, starting at 0.001 and decaying every 50 epochs after a 10-epoch warm-up at  $1e-6$ . Training lasts for 2,000 epochs, with the training data split into training and test sets (90/10 ratio) and progressively divided into patches of  $500 \times 500$ ,  $1000 \times 1000$ , and  $1500 \times 1500$  in each stage.

### 3.24. LR-LL

**Description:** Our solution builds on LLNET [18] for low-light image enhancement in the YUV420 color space, as shown in Fig. 17. The approach consists of three key steps: 1) downscaling high-resolution low-light images and using a lightweight CNN to enhance them for fast processing, 2) applying guided upsampling to model the transformation between low-resolution input and output, and 3) using the estimated model to enhance high-resolution images, enabling real-time processing at full resolution. By performing most computations at lower resolution, the model achieves high-quality enhancement while reducing computational costs. Additionally, we introduce a dataset [1] of low-light images with corresponding long-exposure references, captured in real-world conditions using smartphones.

**Implementation:** Implemented in TensorFlow and trained on a single NVIDIA A100 GPU using the NTIRE 2025 low-light enhancement dataset, the model uses the ADAM optimizer with a learning rate of  $1e-4$  for 1,000 epochs. The loss function is  $\mathcal{L} = 2 \cdot \mathcal{L}_1 + \mathcal{L}_{perc}$ , where  $\mathcal{L}_1$  is the mean absolute error and  $\mathcal{L}_{perc}$  measures feature loss with a trained VGG-19 model. Designed for efficient smartphone deployment, the model can be converted to TFLite, utilizing the GPU delegate for processing. On the Qualcomm Adreno 735, it requires approximately 400MB of memory and processes a  $2000 \times 2992$  image in around 100ms.

### 3.25. X-L

**Description:** Inspired by the SYSU-FVL-T2 approach from the NTIRE-2024 low-light image enhancement challenge [35], we propose a method using ESDNet-L as the backbone. The backbone features an encoder-decoder network with three scales and skip connections, with features generated through bilinear interpolation. At each scale, Semantic-Aligned Scale-Aware Modules are used to enhance scale variation handling, incorporating a pyramid context extraction module and cross-scale dynamic fusion for selective feature fusion. Our modification adds permuted self-attention blocks after the SAM modules, improving the model’s ability to capture global dependencies and refine feature representations. For a detailed illustration, see Fig. 18. The integration of self-attention with SAM improves the handling of scale variations, enhancing the model’s overall performance.

**Implementation:** We adopt a training strategy similar to SYSU-FVL-T2, using a single NVIDIA 4090 GPU.

### 3.26. Team\_IITRPR

**Method description:** Our network, inspired by C2AIR [25], consists of three modules, as illustrated in Fig. 19. The first component, the Degradation-Aware Query Modulated (DAQM) Block, adapts to varying lighting conditions by learning the degradation caused by underexposure. It modulates feature representations using illumination cues to emphasize dark regions needing enhancement. The second module, the Cross Collaborative Feed-Forward (CCF) Block, restores spatial details at multiple scales, ensuring both fine textures and large-scale structures are reconstructed. The third module, the Adaptive Gated Feature Fusion Block (AGFF), selectively integrates features across scales using a gating mechanism, suppressing noise and irrelevant content for naturally enhanced outputs.

**Implementation:** The network is trained on the NTIRE 2025 challenge dataset using an NVIDIA GeForce RTX 1080 GPU with 8 GB of memory and a batch size of 1. The ADAM optimizer is used with a learning rate of  $3 \times 10^{-4}$ ,  $\beta_1 = 0.5$ , and  $\beta_2 = 0.99$ . The model is trained for 110 epochs with random  $512 \times 512$  patches and L1 loss.

### 3.27. CV-SVNIT (Excluded from Report)

### 3.28. JHC-INFO (Excluded from Ranking)

**Description.** RetinexRWKV, as shown in Figs. 20 and 21, is a lightweight model for low-light image enhancement, based on Retinexformer and RWKV TimeMix as shown in Fig. 22. It efficiently integrates spatial and temporal information, supports up to 8K resolution with minimal computational load, and features linear attention with  $O(n)$  complexity for long-range dependencies. Its dynamic state mechanism (RWKV v7) boosts adaptability and generalization across various visual enhancement tasks. More information can be found in [29].

**Implementation.** The model was trained using an AMD Radeon Pro W7900 GPU (48GB VRAM) with Triton acceleration in ROCm, showing strong hardware compatibility. However, pre-trained weights may not be reproducible on Nvidia systems, though training, forwarding, and testing remain feasible. With a batch size of 8, each epoch took 50 seconds, completing 100 epochs on the NTIRE 2025 dataset in 1.5 hours. To improve generalization and training speed,  $256 \times 256$  random cropping was applied in batch training.

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